

Workbook: The Integumentary System

Week 1 workbook · BIO 004 Human Anatomy · Summer 2026

A comprehensive workbook on the skin and its accessory structures: the layers of the epidermis, the cells of the epidermis, the dermis and hypodermis, hair, nails, and glands. Includes burns and skin imaging as clinical-anatomy reference points. Basic functions are included; complex molecular signaling is not.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

The integument overview · 3 questions

Q01 · DOK 1 · Recall

List the major functions of the integumentary system.

Q02 · DOK 1 · Recall

Name the major layers of the integument from outside to inside.

Q03 · DOK 2 · Compare

Compare thick skin and thin skin by where each is found and how their epidermal layers differ.

The epidermis: layers · 7 questions

Q04 · DOK 1 · Recall

Name the strata of the epidermis from deep to superficial.

Q05 · DOK 1 · Recall

Describe the stratum basale, including the cells found there.

Q06 · DOK 1 · Recall

Describe the stratum spinosum and explain its name.

Q07 · DOK 1 · Recall

Describe the stratum granulosum and what happens to keratinocytes there.

Q08 · DOK 1 · Recall

Describe the stratum lucidum and where it occurs.

Q09 · DOK 1 · Recall

Describe the stratum corneum.

Q10 · DOK 2 · Apply

Trace the life of a keratinocyte from its origin to its loss from the body.

The epidermis: cell types · 5 questions**Q11 · DOK 1 · Recall**

What is a keratinocyte and what does it produce?

Q12 · DOK 1 · Recall

What is a melanocyte, where is it found, and what does it do?

Q13 · DOK 1 · Recall

What is a Langerhans (dendritic) cell, where is it found, and what does it do?

Q14 · DOK 1 · Recall

What is a Merkel cell, where is it found, and what does it do?

Q15 · DOK 2 · Compare

Three factors shape natural skin color. Name them and explain how each contributes.

The dermis and hypodermis · 5 questions**Q16 · DOK 1 · Recall**

Name the two layers of the dermis and the kind of connective tissue in each.

Q17 · DOK 2 · Apply

What are dermal papillae, what do they create at the skin surface, and what do they do for the avascular epidermis?

Q18 · DOK 1 · Recall

List the structures that live in the dermis.

Q19 · DOK 1 · Recall

What is the hypodermis, what is it made of, and what does it do?

Q20 · DOK 2 · Apply

Stretch marks (striae) appear in skin that has been rapidly stretched. Which layer is responsible, and why do the marks remain even after the stretch is over?

Hair · 4 questions**Q21 · DOK 1 · Recall**

Name the major parts of a hair and its follicle.

Q22 · DOK 1 · Recall

What is hair made of and how does its keratin differ from that of the skin?

Q23 · DOK 1 · Recall

What determines hair color, and what produces gray hair?

Q24 · DOK 1 · Recall

What is the arrector pili, what kind of structure is it, and what does it do?

Nails · 2 questions

Q25 • DOK 1 • Recall

Name the major parts of a fingernail.

Q26 • DOK 1 • Recall

Where do nails grow from, and what tissue makes them?

Skin glands • 4 questions**Q27 • DOK 1 • Recall**

Describe sebaceous glands: where they sit, what they secrete, by what mechanism, and onto what.

Q28 • DOK 1 • Recall

Describe eccrine sweat glands and what they do.

Q29 • DOK 1 • Recall

Describe apocrine sweat glands and how they differ from eccrine glands.

Q30 • DOK 1 • Recall

What are ceruminous and mammary glands, and what kind of gland is each?

Q31 · DOK 1 · Recall